

Mohamed Abdelrahman

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EDUCATION

University of Houston

Bachelor of Science in Computer Science - Major GPA: 3.80

Expected Graduation: Dec 2026

Houston, TX

EXPERIENCE

Course Facilitator (Incoming – Spring 2026)

Jan 2026 – May 2026

Department of Computer Science - University of Houston

Houston, TX

- Preparing coding problems and instructional material to facilitate COSC 3320: Algorithms and Data Structures for undergraduate students

Software Engineering Intern

Aug 2022 – Jan 2023

bld.ai

Remote

- Developed a full-stack platform for matching families for playdate arrangements using React and Django
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month, following a two-week sprint cycle
- Adhered to Agile Practices with sprint planning, daily stand-ups, and retrospectives optimizing workflow by 50%

LEADERSHIP

Director of Coding Competitions

Jan 2025 – Present

IEEE-NSM - University of Houston

- Prepared and hosted workshops teaching data structures and algorithms in C++, and curated competitive programming competitions to strengthen algorithmic problem-solving skills
- Coordinated with students and faculty regarding ICPC hosting logistics at UH, organizing scheduling shifts for 20 volunteers to ensure successful contest execution
- Organized practice sessions for ICPC team by collecting problems and arranging virtual contest sessions, contributing to securing 1st place at UH Site

Code[Coogs] - University of Houston

- Organized and hosted 9 competitive programming events on Kattis, curating problem sets focused on data structures, algorithms, and complexity analysis for 30+ students

PROJECTS

TallyCart | *Django, EasyOCR, Prompt Engineering*

- Designing Django backend with relational data models to structure product data for optimized shopping recommendations from receipt-derived information
- Integrated EasyOCR with LLM-powered postprocessing to convert raw receipt text into structured product records achieving 98% average accuracy

C++ Ordered Map Library | *C++17, CMake, Catch2, GitHub Actions, Test-Driven Development*

- Developed high-performance ordered map data structure combining doubly linked list with hash table for O(1) lookups while preserving insertion order, with STL-compatible interface and exception safety
- Implemented comprehensive test suite achieving 97.3% code coverage across 2,246 test assertions with Catch2, established CI/CD pipeline via GitHub Actions, and verified zero memory leaks through Valgrind analysis

Nota | *Flutter, Dart, Bloc, SQFlite, Quill, Material 3*

- Developed rich-text note-taking app for Android with offline storage using SQFlite, integrated Flutter Quill editor for formatting, and implemented labeling and filtering features for efficient organization
- Managed UI and business logic with Bloc state management and designed responsive UI with Material 3 components supporting light/dark theme switching

AWARDS

Golden Medalist, ICPC South Regional Contest - UH Site

Nov 2025

University of Houston

- Secured 1st place victory at UH Site and ranked 16th overall among competing teams in the competitive South Regional Contest by solving 4 problems using C++ and algorithmic techniques

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart, C#

Frameworks: Django, React, NextJS, Flutter, Unity

Tools: Git, Docker, Postman, Coverage, GitHub Actions, VS Code, Ubuntu Linux, Bash